Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/15/2025
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Augmented-Program for Levitation Utilization in Space (A-PLUS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/01/2023	**End Date:**	01/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nawer, Jannatun (Tufts University)		
Grant/Contract No.:	80NSSC23K0573 P00002		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project is a continuation of the previous “Round Robin Thermophysical Property measurement” which supports thermophysical property testing on metallic melts using the JAXA ELF and NASA/MSFC electrostatic levitation facilities. This NASA grant directly supports experiments that were conducted using the JAXA ISS-ELF facility onboard the International Space Station and ground-based testing at the NASA/MSFC ESL facility to complement ongoing investigations being performed on another grant using parabolic flight TEMPUS-EML. This report includes results from the study of four key thermophysical properties on industrial superalloys CMSX-4 Plus and Inconel 625. Density and thermal expansion coefficient have been measured as paired properties for both superalloys in JAXA ISS-ELF. Surface tension and viscosity of CMSX-4 Plus was successfully studied using measured successfully in NASA/MSFC ESL facility using droplet oscillation method. Additional testing was conducted using TEMPUS EML facility to study surface tension and viscosity using Science Coupling Electronics (SCE) method. An attempt has been made to quantify the possible sources of uncertainties associated with each measured property during this project using the GUM (Guide to		

	the Expression of Uncertainty in Measurement) principles. This enables future researchers to assess the reliability of the reported values and better understand the advantages and challenges associated with each facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	(i) Disseminate an archived baseline dataset allowing researchers to validate models which account for gravity-induced systematic error in property measurement. Industry will be better able to correlate ground and space results to expand the quantity and quality of available information without the requirement for access to space. (ii) Advance the understanding of how to best allocate the scant resources available for space-based materials property measurement using levitation. In the future this could lead to development of a model where fidelity and cost can be readily calculated for a given material classification and serve as the basis for prioritization of specific ground and space thermophysical property investigations. We will answer the basic question “Can you justify going to space?”
Task Progress:	The ISS-ELF facility was initially designed to process high temperature oxides, and through this study it was shown that this facility could also process a varied range of superalloys. Two industrial superalloys, CMSX-4 Plus and Inconel 625, were successfully levitated and melted at this facility. [Ed. Note: The Japanese Space Agency's electrostatic levitation furnace (ELF) is located on the International Space Station (ISS).] Material Specific Conclusions CMSX-4 Plus • Several CMSX-4 Plus have been processed at the NASA Marshall Space Flight Center (MSFC) in a vacuum environment for a wide temperature range. • Langmuir’s equation, combined with the activity coefficient from Thermo-Calc, provided a prediction of mass loss within an error of 1% for the samples run at NASA MSFC. This evaporation model was also later implemented on the CMSX-4 Plus samples run at ISS-ELF for an initial prediction. • The density of this material has been measured at both space and ground-based facilities and they are both in good agreement. • The surface tension and viscosity of CMSX-4 Plus has been reported for the first time. These are both comparable to another published CMSX alloy. Inconel 625 • Inconel 625 has been successfully levitated and melted on the space-based ISS-ELF and TEMPUS EML (electromagnetic levitation) parabolic flight, both in an Argon gas environment. • ISS-ELF samples heavily oxidized right after the melt and subsequent outgassing was observed. • The density measured from one of the samples at ISS-ELF shows good agreement with a published literature value. • Molybdenum (Mo) is present in this alloy system, which is among the oxidizing elements in the Ellingham diagram. It will thus oxidize first, which does not turn liquid, but rather sublimates. This vapor may have been visible during two-sample processing. • Surface tension of Inconel 625 has been measured using the Sample Coupling Electronics (SCE) device at the TEMPUS EML parabolic flight experiments, which are comparable to a ground-based EML and electrostatic levitator (ESL) facility. • The viscosity of Inconel 625 has also been reported for the first time. To summarize this report, the thermophysical property evaluation will obviously be governed by the reliability of the experimental data on which it is based, and through this study an attempt has been made to provide a meticulous evaluation of the uncertainty. The ISS-ELF facility was initially designed to process high temperature refractory materials, and this study has shown that this facility could not only process metallic samples over a wide temperature range but also can produce high quality thermophysical property data for superalloys. The TEMPUS EML facility exhibited excellent temperature control and sample stability due to larger sample size and higher positioning force.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Nawer J, Ishikawa T, Oda H, Saruwatari H, Koyama C, Xiao X, Schneider S, Kolbe M, Matson DM. "Uncertainty analysis and performance evaluation of thermophysical property measurement of liquid Au in microgravity." npj Microgravity. 2023 May 24;9(1):38. https://doi.org/10.1038/s41526-023-00277-0 , May-2023
Articles in Peer-reviewed Journals	Nawer J, Ishikawa T, Oda H, Koyama C, Saruwatari H, Sansoucie MP, Philips B, Gosse S, Mohr M, Kolbe M, Matson DM. "A quantitative comparison of thermophysical property measurement of CMSX-4® Plus (SLS) in microgravity and terrestrial environments." High Temperatures-High Pressures. 2023 May 1. http://dx.doi.org/10.32908/hthp.v52.1407 , May-2023
Articles in Peer-reviewed Journals	Nawer J, Ishikawa T, Oda H, Koyama C, Matson DM. "Uncertainty quantification of thermophysical property measurement in space and on Earth: A Study of liquid platinum using electrostatic levitation." J. Astron. Space Sci. 2023 Sep 15; 40(3):93-100. https://doi.org/10.5140/JASS.2023.40.3.93 , Sep-2023
Articles in Peer-reviewed Journals	Nawer J, Stanford B, Kolbe M, Schneider S, Gossé S, Wunderlich RK, Mohr M, Borzì A, Neels A, Matson DM. "Thermodynamic assessment of evaporation during molten steel testing onboard the International Space Station." npj Microgravity. 2024 Jul 19;10(1):77. https://doi.org/10.1038/s41526-024-00416-1 , Jul-2024